14.TCP/IP suite:

1. Create a side-by-side comparison table showing at least 4 key differences between the OSI Model and the TCP/IP Model, including the number of layers, layer names, and where protocols like HTTP, TCP, and IP fit.

ANS:

OSI Model vs TCP/IP Model

OSI Model	TCP/IP Model
1. Number of layers: 7 layers	1. Number of layers: 4 layers
2. Layer names: Application, Presentation, Session, Transport, Network, Data Link, Physical	2. Layer names: Application, Transport, Internet, Network Access
3. Application protocols: HTTP works at the Application layer	3. Application protocols: HTTP works at the Application layer
4. Transport protocol: TCP works at the Transport layer	4. Transport protocol: TCP works at the Transport layer
5. Network protocol: IP works at the Network layer	5. Network protocol: IP works at the Internet layer
6. Data Link & Physical: Separate layers	6. Network Access: Combines Data Link and Physical functions
7. Usage: Mainly used as a reference/teaching model to understand networking	7. Usage: Practical protocol suite used as the foundation of the Internet

Layer Mapping

Application TCP/IP (4 Layers)

Application	Application
Presentation	Application

Application	TCP/IP (4 Layers)
Session	application
Transport	Transport
Network	Internet
Data Link	Network Access
Physical	Network Access

Examples:

- HTTP → Application
- TCP → Transport
- IP → Network (OSI) / Internet (TCP/IP)
- Ethernet → Data Link (OSI) / Network Access (TCP/IP)

2. Open Wireshark, start a capture on your active network interface, and visit any website (like www.youtube.com) in your browser. Stop the capture and find a TCP packet in the list. Write down the source IP, destination IP, source port, destination port, and the TCP flag(s) set for that packet.

ANS:

Steps

1. Open Wireshark.
2. Select your active interface, usually Wi-Fi.
3. Click Start Capture.
4. Open your browser and visit www.youtube.com.
5. Wait for the page to load.
6. Stop the capture using the red Stop button.

7. In the filter bar, enter:

tcp

8. Select any TCP packet from the packet list.
9. In the middle Packet Details pane, expand Transmission Control Protocol.

Write these values

Field	Where to find it
Source IP	Internet Protocol → Source
Destination IP	Internet Protocol → Destination
Source Port	TCP → Source Port
Destination Port	TCP → Destination Port
TCP Flags	TCP → Flags

Your answer should look like this:

Source IP: [value from your packet]

Destination IP: [value from your packet]

Source Port: [value from your packet]

Destination Port: [value from your packet]

TCP Flags: [e.g., ACK, PSH, SYN, FIN]

Tip: If you select the first packet of a TCP connection, the flags may be SYN. Other packets commonly show ACK, PSH, ACK, or FIN, ACK.

3. In Wireshark, filter packets by 'udp' and find a DNS query. Identify and note which TCP/IP layer each of the following fields belongs to: source IP, destination IP, source port, destination port, DNS query name.

ANS:

Wireshark DNS Query

1. Open your Wireshark capture.
2. In the filter bar, type:

   udp
3. Press Enter.
4. Find a packet whose Protocol is DNS.
5. Select it and expand:
   - Internet Protocol Version 4 (IPv4)
   - User Datagram Protocol (UDP)
   - Domain Name System (query)

Answer

Field	TCP/IP Layer	Protocol
Source IP	Internet Layer	IP
Destination IP	Internet Layer	IP
Source Port	Transport Layer	UDP
Destination Port	Transport Layer	UDP
DNS Query Name	Application Layer	DNS

Example

For a DNS query such as www.example.com:

Source IP: 192.168.1.5 → Internet Layer

Destination IP: 8.8.8.8 → Internet Layer

Source Port: 54321 → Transport Layer

Destination Port: 53 → Transport Layer

DNS Query Name: www.example.com → Application Layer

**4. Analyze any captured IPv6 packet in Wireshark and list the differences you observe in the IP header compared to an IPv4 packet.

Hint: Look for differences in address format, header length, and any extra fields present in IPv6.**

ANS:

IPv4 vs IPv6 Header Differences in Wireshark

Select an IPv6 packet in Wireshark and expand Internet Protocol Version 6. Compare it with an IPv4 packet.

Feature	IPv4	IPv6
Address format	32-bit address, e.g. 192.168.1.10	128-bit address, e.g. 2001:db8::1
Header length	Variable: 20–60 bytes	Fixed: 40 bytes
Address fields	Source and Destination IPv4 addresses	Source and Destination IPv6 addresses
Header length field	Has IHL (Internet Header Length)	No IHL field because the header has a fixed length
TTL	TTL (Time to Live)	Hop Limit
Protocol field	Protocol identifies TCP, UDP, etc.	Next Header identifies the next protocol/header

Feature	IPv4	IPv6
Fragmentation	Fragmentation information can be in the main IPv4 header	Fragmentation uses a separate Fragment Extension Header
Checksum	IPv4 header has a Header Checksum	IPv6 has no header checksum

Main observations for your practical

IPv6 has:

- Much longer 128-bit addresses
- A fixed 40-byte base header
- No IHL
- Hop Limit instead of TTL
- Next Header instead of Protocol
- No IPv6 header checksum
- Optional extension headers for additional information